

Disruption-Aware Logistics Planning

UNDERSTAND THE PROBLEM

Context

Logistics operations depend on schedules, routes, vehicles, delivery requirements, and many interconnected activities. Disruptions such as road closures, accidents, severe weather, vehicle problems, transport delays, changed delivery requirements, or unexpected constraints can quickly affect an existing plan.

Problem

When a disruption occurs, logistics operators may struggle to understand what has changed, which deliveries or routes are affected, what dependencies are involved, and what should happen next. One change can affect several connected activities, and limited visibility can result in significant time spent identifying the impact and deciding how to respond.

Objective

Understand the challenges logistics teams face when responding to changing conditions and develop software that helps them make informed and timely decisions.

Target Users

- Logistics operators.

- Transportation teams.

- Delivery management teams.

- Operations teams responsible for coordinating logistics activities.

Disruption-Aware Logistics Planning

— EMPATHY & USER NEEDS

Logistics Operators

Understand what changed and why it matters.

Identify deliveries affected by a disruption.

Understand route and schedule impacts.

See important dependencies between activities.

Recognize which situations need immediate attention.

Coordinate changes and make decisions without unnecessary delay.

Adapt plans when the original plan is no longer suitable.

Delivery & Transportation Teams

- Understand vehicle and route availability.

- Know when delivery plans have changed.

- Identify activities that require attention.

- Coordinate changes across the operation.

- Reduce avoidable delays.

- Respond effectively when conditions change.

Common Pain Points

- Unexpected disruptions.

- Multiple deliveries and connected activities.

- Changing road and transport conditions.

- Vehicle availability problems.

- Customer or delivery requirement changes.

- Limited visibility of dependencies.

- Difficulty identifying priorities.

- Delayed decisions and manual coordination.

- Difficulty adapting the original plan.

Disruption-Aware Logistics Planning

BUILD FOR CHANGING CONDITIONS

Build software that helps logistics teams respond when plans are disrupted.

Help users answer three important questions:

What changed?



What is affected?



What should they do next?

Think About

- How can disruptions become visible quickly?
- How can dependencies between activities be made easier to understand?
- How can possible actions be considered when conditions change?
- How can affected deliveries, routes, or activities be understood?
- How can urgent situations be identified?
- How can plans be adapted without adding unnecessary complexity?

Your Approach

Decide what your product should help users understand and what actions it should support. The technology, features, and implementation approach are open.

Build something that helps logistics teams stay in control when plans change.